

List of publication and products with abstracts

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Via, a 14 h-index, a 16 i10-index, and total citations: 4002 ([Google Scholar 2026](#)), my most important work is represented in the following papers:

Accepted

- [1] P. Yovkochev and **Ntelis, P.**, 2026. Evolution and Mass Decay of a Gaussian Pulse in Time-Fractional Diffusion: An L1-Implicit Scheme Analysis. *Turanian Journal*, vol. 2, no. 1, pp. 020103-1–020103-7. [URL](#), Contribution: Supervision, and co-design of analysis. (Citations: 0)
Abstract: We apply the L1-implicit finite difference scheme to solve the time-fractional diffusion equation with a Gaussian initial condition. The Caputo fractional derivative of order $\alpha \in (0, 1]$ is discretized using the L1 approximation on a uniform temporal mesh, while the spatial second derivative is approximated by a central difference scheme. Numerical simulations are performed to study the effect of the fractional order α on the solution profile and to investigate mass conservation properties. The results demonstrate the characteristic subdiffusive behavior and the non-conservation of mass typically observed in fractional diffusion processes.
- [2] **Ntelis, P.** 2025. Advancing Tensor Theories. Symmetry: Mathematics: Advances in Topology and Algebraic Geometry. [URL](#) (Citations: 3) ★
Abstract: Advancing Tensor Theories (ATT); This paper advances the foundations of tensor and category theories by introducing novel concepts and rigorous constructive proofs. We generalize tensor theory through the innovative notion of a generalized tensor index, a versatile framework that unifies diverse tensor indices, and explore its transformation properties. Using fractional derivatives, we provide a geometrical interpretation of these generalized tensors, revealing new insights into its structure. Additionally, we forge a deep connection between tensor and category theories, integrating sets, tensors, categories, and functors with extensions like partial differentiation and integration. This synthesis yields original constructs—setorial tensors, categorial tensors, and functorial tensors—which open uncharted pathways in mathematical analysis. Our contributions not only extend prior research but also significantly enhance tensor theory, category theory, set theory, logic, topology, algebraic geometry, foundations, and philosophy, with potential applications spanning physics, geometry, and beyond.
- [3] **Ntelis, P.** 2025. Advanced manifold-metric pairs. Mathematics. [ArXiv](#), [URL](#) (Citations: 1) ★
Abstract: Advanced manifold-metric pairs. (AMMP); This article presents a novel mathematical formalism for advanced manifold–metric pairs, enhancing the frameworks of geometry and topology. We construct various D-dimensional manifolds and their associated metric spaces using functional methods, with a focus on integrating concepts from mathematical physics, field theory, topology, algebra, probability, and statistics. Our methodology employs rigorous mathematical construction proofs and logical foundations to develop generalized manifold–metric pairs, including homogeneous and isotropic expanding manifolds, as well as probabilistic and entropic variants. Key results include the establishment of metrizability for topological manifolds via the Urysohn Metrization Theorem, the formulation of higher-rank tensor metrics, and the exploration of complex and quaternionic codomains with applications to cosmological models like the expanding spacetime. By combining spacetime generalized sets with information-theoretic and probabilistic approaches, we achieve a unified framework that advances the understanding of manifold–metric interactions and their physical implications.

- [4] **Ntelis, P.** & J. L. Said, 2025. Analytical $poly\Lambda$ CDM dynamics. PDU.[URL](#) (Citations: 1) ★
Abstract: In this paper, we study the ϕ CDM and the $poly\Lambda$ CDM models and compare them to the standard concordance model, the vanilla Λ CDM. Analytical solutions for the ϕ CDM, vital for precision, show sub-% agreement with Λ CDM due to selected energy density components, yet larger differences from numerical solutions of the same equations underscore their reliability over numerical integration of stiff systems. The $poly\Lambda$ CDM model is a phenomenological modified gravity model. We analyze their respective epoch evolutions, perform a detailed analytical and numerical dynamical analysis for each, and conduct a comparative study between the frameworks. Our work revitalizes these models by incorporating systems with an expanded set of variables, enabling us to derive analytical expressions for the energy density ratios of all species in both models. Moreover, the $poly\Lambda$ CDM model we present is both more comprehensive and streamlined compared to existing studies. It captures all known cosmic epochs, including radiation, matter, and dark energy phases, and more exotic ones. The $poly\Lambda$ CDM model has the cosmological constant term dominance in the far future epoch. Our findings show that all three models effectively describe the widely accepted cosmic evolution scenario and align with current observations. While all Λ CDM, ϕ CDM, and $poly\Lambda$ CDM reproduce the qualitative and quantitative behavior of the universe's species across epochs, the $poly\Lambda$ CDM model exhibits a richer phenomenology, offering deeper insights into cosmic evolution.
- [5] E.D. Valentino, J. L. Said, ..., **Ntelis, P.** ..., et al. 2025. The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics. Contribution: Writing on extended theories of gravity, scientific review and contribution on discussions for further research development. PDU [URL](#) (Citations: 472) *Abstract:* The standard model of cosmology has provided a good phenomenological description of a wide range of observations both at astrophysical and cosmological scales for several decades. This concordance model is constructed by a universal cosmological constant and supported by a matter sector described by the standard model of particle physics and a cold dark matter contribution, as well as very early-time inflationary physics, and underpinned by gravitation through general relativity. There have always been open questions about the soundness of the foundations of the standard model. However, recent years have shown that there may also be questions from the observational sector with the emergence of differences between certain cosmological probes. In this White Paper, we identify the key objectives that need to be addressed over the coming decade together with the core science projects that aim to meet these challenges. These discordances primarily rest on the divergence in the measurement of core cosmological parameters with varying levels of statistical confidence. These possible statistical tensions may be partially accounted for by systematics in various measurements or cosmological probes but there is also a growing indication of potential new physics beyond the standard model. After reviewing the principal probes used in the measurement of cosmological parameters, as well as potential systematics, we discuss the most promising array of potential new physics that may be observable in upcoming surveys. We also discuss the growing set of novel data analysis approaches that go beyond traditional methods to test physical models. These new methods will become increasingly important in the coming years as the volume of survey data continues to increase, and as the degeneracy between predictions of different physical models grows. There are several perspectives on the divergences between the values of cosmological parameters, such as the model-independent probes in the late Universe and model-dependent measurements in the early Universe, which we cover at length. The White Paper closes with a number of recommendations for the community to focus on for the upcoming decade of observational cosmology, statistical data analysis, and fundamental physics developments.
- [6] Ntelis, P. 2022. A (D_τ, D_x) -manifold with N -correlators of N_t -objects. IJGMMP, arXiv e-prints arXiv:2209.07472 [URL](#), (Citations: 1)
Abstract: Context. In this paper, we describe a mathematical formalism for a (D_τ, D_x) -dimensional manifold with N -correlators of N_t types of targeted source objects, with cross correlations and contaminants. Methodology. In particular, we build this formalism using simple notions of mathematical physics, field theory, topology, algebra, statistics n-correlators and Fourier transform.

Results. We present and discuss the applicability of this formalism in the context of cosmological scales, i.e. from astronomical scales to quantum scales, for which we give some intuitive examples, explicitly, for standard spacetime dimensions and extra dimensions. Conclusion. We conclude that this study can be used as a guide to analyse and build models for future cosmological and collider experiments. Furthermore, this study opens the road of extra dimension studies.

- [7] **Ntelis, P.** & J. L. Said 2025. Exploring ϕ CDM model dynamics. EPJC, arXiv:2502.03486v1 [URL](#), (Citations: 4)

Abstract: In this paper, we introduce the ϕ CDM model, and we compare to the concordance model. We provide their epoch evolution, we perform a dynamical analysis to each model, and a comparative analysis between the two models. We revitalise these two models, since we study them in systems with higher number of variables. Furthermore, the ϕ CDM model is more complete than the one that we found in the literature, since we take into account all the so far discovered epochs, radiation, matter, and dark energy epochs. In contrast in the literature there is no study including the radiation epoch for this model. We find that both models can describe the generally accepted scenario of cosmic evolution, and current observations. Both models, describe qualitative and quantitative current observations about the epoch behaviour of the species of the universe. We find the ϕ CDM model has the following exotic transition from a *dominant radiation energy density ratio* epoch and a *low dominant matter energy density ratio and high dominant radiation energy density ratio* epoch, towards a *dominant scalar potential energy density ratio* epoch, in the matter-potential-scalar plane.

- [8] **Ntelis, P.** & J. L. Said 2025. Simple $\phi\Lambda$ CDM dynamics. IJGMMP, [URL](#), (Citations: 3)

Abstract: In this paper, we introduce the $\phi\Lambda$ CDM model and compare it to the concordance model. We present their respective epoch evolutions, perform a detailed dynamical analysis for each model, and conduct a comparative analysis between the two. This study revitalizes these models by considering systems with a higher number of variables. Additionally, the $\phi\Lambda$ CDM model we present is both more comprehensive and simpler than those found in the literature, as it accounts for all known epochs, including the radiation, matter, and dark energy phases. Notably, existing studies on this model often omit the radiation epoch and focus on simpler dynamics. We find that both models, the $\phi\Lambda$ CDM and the Λ CDM, can describe the generally accepted scenario of cosmic evolution, and current observations. Both models, describe qualitative and quantitative current observations about the epoch behaviour of the species of the universe. We find the $\phi\Lambda$ CDM model has the following exotic transition from a *dominant radiation energy density ratio* epoch, or a *low scalar kinetic term energy density ratio* epoch, towards a *dominant cosmological constant energy density ratio* epoch. This renders the $\phi\Lambda$ CDM a richer phenomenologically model than the concordance cosmological model. The software of the study is publicly available [SimplePhiLambdaCDM](#).

- [9] S.Casas, ..., **Ntelis, P.**, and 128 colleagues 2025. Euclid: Constraints on $f(R)$ cosmologies from the spectroscopic and photometric primary probes. A&A, [URL](#) (Citations: 27) Contribution: Scientific review, software development.

Abstract: Euclid will provide a powerful compilation of data including spectroscopic redshifts, the angular clustering of galaxies, weak lensing cosmic shear, and the cross-correlation of these last two photometric observables. This will lead to very stringent constraints on the Λ CDM concordance cosmological model and models beyond it, in which for instance standard gravity is modified. In this study we extend recently presented Euclid forecasts into the Hu-Sawicki $f(R)$ cosmological model, a popular extension of the Hilbert-Einstein action that introduces an universal modified gravity force in a scale-dependent way. This scale-dependent modification requires a generalisation of our previous recipes for spectroscopic and photometric galaxy clustering, and for weak lensing, both in the linear and in the non-linear regimes. Our aim is to estimate how well future Euclid data will be able to constrain the extra parameter of the theory, f_{R0} , for the range in which this parameter is still allowed by current observations. For the spectroscopic probe, we use a phenomenological approach to account for the scale dependence of the growth of perturbations in the terms related to baryon acoustic oscillations and redshift-space distortions. For the photometric observables, which probe deeper into the non-linear regime, we use a fitting formula developed in the literature that

captures the modifications in the non-linear matter power spectrum caused by the $f(R)$ model. We show that, in an optimistic setting, and for a fiducial value of $f_{R0} = 5 \times 10^{-6}$, Euclid alone will be able to constrain the additional parameter $\log f_{R0}$ at the 3% level, using spectroscopic galaxy clustering alone; at the 1.4% level, using the combination of photometric probes on their own; and at the 1% level, using the combination of spectroscopic and photometric observations. This last constraint corresponds to an error of the order of 6×10^{-7} at the 1σ level on the model parameter $f_{R0} = 5 \times 10^{-6}$. We report also forecasted constraints on a model with a large value of f_{R0} , namely $f_{R0} = 5 \times 10^{-5}$ and a model closer to standard gravity with $f_{R0} = 5 \times 10^{-7}$ and show that in the optimistic scenario, Euclid will be able to distinguish these models from Λ CDM at more than 3σ . We find that with a good control of systematic effects and modelling of the matter power spectrum in the mildly and deeply non-linear regime, Euclid will be a powerful probe for $f(R)$ models. It will constrain the scale-dependence of the perturbations as well as a substantial part of the non-linear regime of structure formation, discerning these models from Λ CDM.

- [10] Euclid Collaboration, and 1137 colleagues 2024. Euclid. I. Overview of the Euclid mission. A&A, arXiv:2405.13491 [URL](#), (Citations: 785) Contribution: Scientific review and contribution on discussions for further research.

Abstract: The current standard model of cosmology successfully describes a variety of measurements, but the nature of its main ingredients, dark matter and dark energy, remains unknown. Euclid is a medium-class mission in the Cosmic Vision 2015–2025 programme of the European Space Agency (ESA) that will provide high-resolution optical imaging, as well as near-infrared imaging and spectroscopy, over about $14\,000\text{ deg}^2$ of extragalactic sky. In addition to accurate weak lensing and clustering measurements that probe structure formation over half of the age of the Universe, its primary probes for cosmology, these exquisite data will enable a wide range of science. This paper provides a high-level overview of the mission, summarising the survey characteristics, the various data-processing steps, and data products. We also highlight the main science objectives and expected performance.

- [11] **Ntelis, P.** & Morris, A. 2023. Functors of actions. Foundations of Physics, arXiv:2010.06707 [URL](#), (Citations: 24) ★

Abstract: Functors of Actions Theories (FAT). In this document, we introduce a novel formalism for any field theory and apply it to the effective field theories of large-scale structure. The new formalism is based on *functors of actions* composing those theories. This new formalism predicts the *actionic* fields. We discuss our findings in a *cosmological gravitology* framework. We present these results with a cosmological inference approach and give guidelines on how we can choose the best candidate between those models with some latest understanding of model selection using Bayesian inference.

- [12] Göksel Karaçaylı, N., ... **Ntelis, P.**, and 10 colleagues 2022. Optimal 1D Ly α Forest Power Spectrum Estimation – II. KODIAQ, SQUAD & XQ-100. MNRAS [URL](#), (Citations: 58), Contribution: Scientific review and contribution on discussions for further research and software development.

Abstract: We measure the 1D Ly α power spectrum P_{1D} from Keck Observatory Database of Ionized Absorption toward Quasars (KODIAQ), The Spectral Quasar Absorption Database (SQUAD) and XQ-100 quasars using the optimal quadratic estimator. We combine KODIAQ and SQUAD at the spectrum level, but perform a separate XQ-100 estimation to control its large resolution corrections in check. Our final analysis measures P_{1D} at scales $k < 0.1\text{ s km}^{-1}$ between redshifts $z = 2.0 - 4.6$ using 538 quasars. This sample provides the largest number of high-resolution, high-S/N observations; and combined with the power of optimal estimator it provides exceptional precision at small scales. These small-scale modes ($k \gtrsim 0.02\text{ s km}^{-1}$), unavailable in Sloan Digital Sky Survey (SDSS) and Dark Energy Spectroscopic Instrument (DESI) analyses, are sensitive to the thermal state and reionization history of the intergalactic medium, as well as the nature of dark matter. As an example, a simple Fisher forecast analysis estimates that our results can improve small-scale cut off sensitivity by more than a factor of 2.

- [13] Hamaus, N., ... **Ntelis, P.**, and 128 colleagues 2022. Euclid: Forecasts from redshift-space

distortions and the Alcock-Paczynski test with cosmic voids. A&A arXiv e-prints [URL](#), (Citations: 95), Contribution: Scientific review and contribution on discussions for further research and software development.

Abstract: Euclid is poised to survey galaxies across a cosmological volume of unprecedented size, providing observations of more than a billion objects distributed over a third of the full sky. Approximately 20 million of these galaxies will have their spectroscopy available, allowing us to map the three-dimensional large-scale structure of the Universe in great detail. This paper investigates prospects for the detection of cosmic voids therein and the unique benefit they provide for cosmological studies. In particular, we study the imprints of dynamic (redshift-space) and geometric (Alcock–Paczynski) distortions of average void shapes and their constraining power on the growth of structure and cosmological distance ratios. To this end, we made use of the Flagship mock catalog, a state-of-the-art simulation of the data expected to be observed with Euclid. We arranged the data into four adjacent redshift bins, each of which contains about 11 000 voids and we estimated the stacked void-galaxy cross-correlation function in every bin. Fitting a linear-theory model to the data, we obtained constraints on f/b and $D_M H$, where f is the linear growth rate of density fluctuations, b the galaxy bias, D_M the comoving angular diameter distance, and H the Hubble rate. In addition, we marginalized over two nuisance parameters included in our model to account for unknown systematic effects in the analysis. With this approach, Euclid will be able to reach a relative precision of about 4% on measurements of f/b and 0.5% on $D_M H$ in each redshift bin. Better modeling or calibration of the nuisance parameters may further increase this precision to 1% and 0.4%, respectively. Our results show that the exploitation of cosmic voids in Euclid will provide competitive constraints on cosmology even as a stand-alone probe. For example, the equation-of-state parameter, w , for dark energy will be measured with a precision of about 10%, consistent with previous more approximate forecasts.

- [14] Ilić, S., ... **Ntelis, P.**, and 207 colleagues 2021. *Euclid* preparation: XV. Forecasting cosmological constraints for the *Euclid* and CMB joint analysis. A&A arXiv e-prints [URL](#), (Citations: 104), Contribution: Scientific review and contribution on discussions for further research. ★

Abstract: The combination and cross-correlation of the upcoming Euclid data with cosmic microwave background (CMB) measurements is a source of great expectation since it will provide the largest lever arm of epochs, ranging from recombination to structure formation across the entire past light cone. In this work, we present forecasts for the joint analysis of Euclid and CMB data on the cosmological parameters of the standard cosmological model and some of its extensions. This work expands and complements the recently published forecasts based on Euclid-specific probes, namely galaxy clustering, weak lensing, and their cross-correlation. With some assumptions on the specifications of current and future CMB experiments, the predicted constraints are obtained from both a standard Fisher formalism and a posterior-fitting approach based on actual CMB data. Compared to a Euclid-only analysis, the addition of CMB data leads to a substantial impact on constraints for all cosmological parameters of the standard Λ -cold-dark-matter model, with improvements reaching up to a factor of ten. For the parameters of extended models, which include a redshift-dependent dark energy equation of state, non-zero curvature, and a phenomenological modification of gravity, improvements can be of the order of two to three, reaching higher than ten in some cases. The results highlight the crucial importance for cosmological constraints of the combination and cross-correlation of Euclid probes with CMB data.

- [15] Alam, S., ... **Ntelis, P.**, and 33 colleagues 2020. Towards testing the theory of gravity with DESI: summary statistics, model predictions and future simulation requirements. JCAP [URL](#), (Citations: 75), Contribution: Scientific review and contribution on discussions for further research and software development. ★

Abstract: Shortly after its discovery, General Relativity (GR) was applied to predict the behavior of our Universe on the largest scales, and later became the foundation of modern cosmology. Its validity has been verified on a range of scales and environments from the Solar system to merging black holes. However, experimental confirmations of GR on cosmological scales have so far lacked the accuracy one would hope for – its applications on those scales being largely based

on extrapolation and its validity there sometimes questioned in the shadow of the discovery of the unexpected cosmic acceleration. Future astronomical instruments surveying the distribution and evolution of galaxies over substantial portions of the observable Universe, such as the Dark Energy Spectroscopic Instrument (DESI), will be able to measure the fingerprints of gravity and their statistical power will allow strong constraints on alternatives to GR.

In this paper, based on a set of N -body simulations and mock galaxy catalogs, we study the predictions of a number of traditional and novel *summary statistics* beyond linear redshift distortions in two well-studied modified gravity models – chameleon $f(R)$ gravity and a braneworld model – and the potential of testing these deviations from GR using DESI. These *summary statistics* employ a wide array of statistical properties of the galaxy and the underlying dark matter field, including two-point and higher-order statistics, environmental dependence, redshift space distortions and weak lensing. We find that they hold promising power for testing GR to unprecedented precision. The major future challenge is to make realistic, simulation-based mock galaxy catalogs for both GR and alternative models to fully exploit the statistic power of the DESI survey (by matching the volumes and galaxy number densities of the mocks to those in the real survey) and to better understand the impact of key systematic effects. Using these, we identify future simulation and analysis needs for gravity tests using DESI.

- [16] Tutusaus, I., ... **Ntelis, P.**, and 89 colleagues 2020. Euclid: The importance of galaxy clustering and weak lensing cross-correlations within the photometric Euclid survey. *Astronomy and Astrophysics* 643, [URL](#), (Citations: 92), Contribution: Scientific review and contribution on discussions for further research.

Abstract: The data from the Euclid mission will enable the measurement of the photometric redshifts, angular positions, and weak lensing shapes for over a billion galaxies. This large dataset, with well controlled systematic effects, will allow for cosmological analyses using the angular clustering of galaxies (GCph) and cosmic shear (WL). For Euclid these two cosmological probes will not be independent because they will probe the same volume of the Universe. The cross-correlation (XC) between these probes can tighten constraints and it is therefore important to quantify their impact for Euclid. In this study we therefore extend the recent Euclid forecasts presented in previous IST paper by carefully quantifying the impact of XC not only on the final parameter constraints for different cosmological models, but also on the nuisance parameters. In particular, we aim at understanding the amount of additional information that XC can provide for parameters encoding systematic effects, such as galaxy bias or intrinsic alignments (IA). We follow the Fisher matrix formalism presented in the IST paper and make use of the codes validated therein. We also investigate a different galaxy bias model, obtained from the Flagship simulation, and additional photometric-redshift uncertainties, and the impact of including the XC terms in constraining these. Starting with a baseline model, we show that the XC terms improve the dark energy Figure of Merit (FoM) by a factor ~ 5 , whilst they also reduce the uncertainties on galaxy bias by $\sim 17\%$ and the uncertainties on IA by a factor ~ 4 . The XC terms also help in constraining the γ parameter for minimal modified gravity models. Concerning galaxy bias, we observe that the role of the XC terms on the final parameter constraints is qualitatively the same irrespective of the specific galaxy bias model used. For IA we show that the XC terms can help in distinguishing between different models, and that if IA terms are neglected then this can lead to significant biases on the cosmological parameters. Finally, we show that the XC terms can lead to a better determination of the mean of the photometric galaxy distributions. We find that the XC between GCph and WL within the Euclid survey is necessary to extract the full information content from the data in future analyses. These terms help in better constraining the cosmological model, and also lead to a better understanding of the systematic effects that contaminate these probes. Furthermore, we find that XC significantly helps in constraining the mean of the photometric-redshift distributions, but, at the same time, it requires a more precise knowledge of this mean, with respect to single probes, in order not to degrade the final FoM.

- [17] **Ntelis, P.**, A. Ealet, S. Escoffier, J.-C. Hamilton, A. J. Hawken, J.-M. Le Goff, J. Rich, and A. Tilquin 2018. The scale of cosmic homogeneity as a standard ruler. *Journal of Cosmology and*

Astroparticle Physics 2018, [URL](#), (Citations: 28)

Abstract: In this paper, we study the characteristic scale of transition to cosmic homogeneity of the universe, $\mathcal{R}_H$, as a standard ruler, to constrain cosmological parameters on mock galaxy catalogues. We use mock galaxy catalogues that simulate the CMASS galaxy sample of the BOSS survey in the redshift range $0.43 \leq z \leq 0.7$. In each redshift bin we obtain the homogeneity scale, defined as the scale at which the universe becomes homogeneous to 1%, i.e. $D_2(\mathcal{R}_H) = 2.97$. With a simple Fisher analysis, we find that the performance of measuring the cosmological parameters with either the position of the BAO peak or the homogeneity scale is comparable. We show that $\mathcal{R}_H$ has a dependence on the galaxy bias. If the accuracy and precision of this bias is achieved to 1%, as expected for future surveys, then $\mathcal{R}_H$ is a competitive standard ruler.

- [18] Blanton, M. R., ... **Ntelis, P.**, and 361 colleagues 2017. Sloan Digital Sky Survey IV: Mapping the Milky Way, Nearby Galaxies, and the Distant Universe. The Astronomical Journal 154, [URL](#), (Citations: 2122), Contribution: Scientific review and contribution on discussions for further research. ★

Abstract: We describe the Sloan Digital Sky Survey IV (SDSS-IV), a project encompassing three major spectroscopic programs. The Apache Point Observatory Galactic Evolution Experiment 2 (APOGEE-2) is observing hundreds of thousands of Milky Way stars at high resolution and high signal-to-noise ratios in the near-infrared. The Mapping Nearby Galaxies at Apache Point Observatory (MaNGA) survey is obtaining spatially resolved spectroscopy for thousands of nearby galaxies (median $z \sim 0.03$). The extended Baryon Oscillation Spectroscopic Survey (eBOSS) is mapping the galaxy, quasar, and neutral gas distributions between $z \sim 0.6$ and 3.5 to constrain cosmology using baryon acoustic oscillations, redshift space distortions, and the shape of the power spectrum. Within eBOSS, we are conducting two major subprograms: the SPectroscopic IDentification of eROSITA Sources (SPIDERS), investigating X-ray AGNs and galaxies in X-ray clusters, and the Time Domain Spectroscopic Survey (TDSS), obtaining spectra of variable sources. All programs use the 2.5 m Sloan Foundation Telescope at the Apache Point Observatory; observations there began in Summer 2014. APOGEE-2 also operates a second near-infrared spectrograph at the 2.5 m du Pont Telescope at Las Campanas Observatory, with observations beginning in early 2017. Observations at both facilities are scheduled to continue through 2020. In keeping with previous SDSS policy, SDSS-IV provides regularly scheduled public data releases; the first one, Data Release 13, was made available in 2016 July.

- [19] **Ntelis, P.**, and 22 colleagues 2017. Exploring cosmic homogeneity with the BOSS DR12 galaxy sample. Journal of Cosmology and Astroparticle Physics 2017, [URL](#), (Citations: 113) ★

Abstract: In this study, we probe the transition to cosmic homogeneity in the Large Scale Structure (LSS) of the Universe using the CMASS galaxy sample of BOSS spectroscopic survey which covers the largest effective volume to date, $3 h^{-3} \text{ Gpc}^3$ at $0.43 \leq z \leq 0.7$. We study the scaled counts-in-spheres, $N(< r)$, and the fractal correlation dimension, $\mathcal{D}_2(r)$, to assess the homogeneity scale of the universe using a *Landy & Szalay* inspired estimator. Defining the scale of transition to homogeneity as the scale at which $\mathcal{D}_2(r)$ reaches 3 within 1%, i.e. $\mathcal{D}_2(r) > 2.97$ for $r > \mathcal{R}_H$, we find $\mathcal{R}_H = (63.3 \pm 0.7) h^{-1} \text{ Mpc}$, in agreement at the percentage level with the predictions of the ΛCDM model $\mathcal{R}_H = 62.0 h^{-1} \text{ Mpc}$. Thanks to the large cosmic depth of the survey, we investigate the redshift evolution of the transition to homogeneity scale and find agreement with the ΛCDM prediction. Finally, we find that $\mathcal{D}_2$ is compatible with 3 at scales larger than $300 h^{-1} \text{ Mpc}$ in all redshift bins. These results consolidate the Cosmological Principle and represent a precise consistency test of the ΛCDM model.

- [20] Laurent, P., ... **Ntelis, P.**, and 17 colleagues 2016. A $14 h^{-3} \text{ Gpc}^3$ study of cosmic homogeneity using BOSS DR12 quasar sample. Journal of Cosmology and Astroparticle Physics 2016, [URL](#), (Citations: 99), Contribution: Scientific review and contribution on discussions for further research and software development.

Abstract: The BOSS quasar sample is used to study cosmic homogeneity with a 3D survey in the redshift range $2.2 < z < 2.8$. We measure the count-in-sphere, $\mathcal{N}(< r)$, i.e. the average number of objects around a given object, and its logarithmic derivative, the fractal correlation dimension,

$D_2(r)$. For a homogeneous distribution $N(< r) \propto r^3$ and $D_2(r) = 3$. Due to the uncertainty on tracer density evolution, 3D surveys can only probe homogeneity up to a redshift dependence, i.e. they probe so-called “spatial isotropy”. Our data demonstrate spatial isotropy of the quasar distribution in the redshift range $2.2 < z < 2.8$ in a model-independent way, independent of any FLRW fiducial cosmology, resulting in $3 - \langle D_2 \rangle < 1.7 \times 10^{-3}$ (2σ) over the range $250 < r < 1200 h^{-1} \text{Mpc}$ for the quasar distribution. If we assume that quasars do not have a bias much less than unity, this implies spatial isotropy of the matter distribution on large scales. Then, combining with the Copernican principle, we finally get homogeneity of the matter distribution on large scales. Alternatively, using a flat ΛCDM fiducial cosmology with CMB-derived parameters, and measuring the quasar bias relative to this ΛCDM model, our data provide a consistency check of the model, in terms of how homogeneous the Universe is on different scales. $D_2(r)$ is found to be compatible with our ΛCDM model on the whole $10 < r < 1200 h^{-1} \text{Mpc}$ range. For the matter distribution we obtain $3 - \langle D_2 \rangle < 5 \times 10^{-5}$ (2σ) over the range $250 < r < 1200 h^{-1} \text{Mpc}$, consistent with homogeneity on large scales.

Under review

- [1] *Ntelis, P.*, et al. 2019. Cosmological constraints from cosmic homogeneity. (Cites: 4) [URL](#)
- [2] *Ntelis, P.* 2025. A probabilistic and expanding Universe. [URL](#), (Cites: 2)
- [3] *Ntelis, P.* 2023. Advancing category and tensor theories. [URL](#), (Cites: 2)
- [4] *Ntelis, P.* 2025. CosmoFATs. [URL](#), (Cites: 2)
- [5] *Ntelis, P.* 2025. CosmoFATs Dynamics. [URL](#) (Cites: 0)
- [6] *Ntelis, P.* 2025. A cosmic Schiff’s conjecture test. [URL](#), , [DOI](#), (Cites: 0)
- [7] *Ntelis, P.* 2025. A universal foundational theory. [URL](#), (Cites: 0)
- [8] *Ntelis, P.* 2026. · A dot. [URL](#), [DOI](#), (Cites: 0)
- [9] *Ntelis, P.* 2026. Analytical sub-poly ΛCDM through cosmic obs. [URL](#), [DOI](#), (Cites: 0)
- [11] *Ntelis, P.* 2026. A foundational analysis of local functional calculus [URL](#), [DOI](#), (Cites: 0)
- [11] *Ntelis, P.*, B.J. Ahmedov, C.Yuan, 2026. A cosmic femtolensing Hawking radiation signature [URL](#), [DOI](#), (Cites: 0)

Universions (Outreach)

Universion is a new compound word from the words universe and the suffix -ion which means action or condition or the word ion which means an atom that is positively or negatively charged. Universion means a being, an object, an action, or a condition of the universe. This channel is dedicated to explaining cosmology the study of the universe as a whole to the general and not-so-general public for all ages. Thanks for your consideration! Enjoy! :)

Outreach	2022- <i>n_{ow}</i> , Universions Youtube Channel
	2025 , Utube, Comet 3i ATLAS: Interstellar Halloween Secrets! (6’)
	2025 , Utube, Are there Aliens ? Drake’s Equation ! (7’)
	2025 , Utube, Quantum Tunnelling in Your Palm! (1’)
	2025 , Utube, Advanced Manifold-Metric Pairs (10’)
	2025 , Utube, A sunny watery magic (33’)
	2025 , Utube, Advanced Euler’s identity (3’)
	2025 , Utube, Analytical polyΛCDM dynamics 101 (5’ talk)
	2025 , Utube, Advancing Tensor Theories 101 (10’ talk)
	2025 , Utube, Quintessence 101 (5’ talk)
	2024 , Utube, Introduction cosmology for everyone (1h talk)
	2023(’24), IBS of Provence , Neuroscience and cosmology for high school students (3h)
	2023 , IBS of Provence , Math and physics from CPT for high school students (3h)
	2022 , IBS of Provence , Mathematics and physics for high school students (3h)

Conferences	2026, HIT, SoP, Advanced math in cosmology (1h Seminar)
Talks/Posters	2026, HIT, SoP, A note on Baye's data analysis (1h Seminar)
Seminars	2025, The Dark Side of the Universe, FAT dynamics to cosmology (25' Talk)
	2025, Cosmo From Home, Advanced math in cosmology (Talk)
	2024, 50 yr of Horndeski at PI, A probabilistic expanding universe under FA , (Poster)
	2024, Cosmo From Home, A probabilistic expanding universe under FA , (25 min talk)
	2023, NEB, New avenues of functors of actions theories , (Talk 20')
	2023, Cosmo From Home, New avenues on Functors of Actions , (25 min talk)
	2023, NTUA HET, New theories in cosmology and particle physics , (1h Seminar)
	2022, CERN TH DPT Visit, Functors of Actions (5' Blackboard Talk)
	2021, Euclid Science Talk, Overview of impact of interlopers on PNG (1h Talk)
	2020, Atelier D.E., PNG with interlopers in Euclid , (20' Talk) HVT
	2020, Atelier D.E., Perturbation Theory of LSS, code: CLASS-PT , (20' Talk)
	2020, SEMFE, NTUA, LSS: Theory and Observations , (40' Seminar)
	2019, 30th SYMPOSIUM, Fractality from Multiple Tracers , (15' Talk)
	2019, Euclid IAS Paris, Systematics on PNG , (20' Talk)
	2019, Euclid Consortium Helsinki, Primordial Non-Gaussianity , (15' Talk)
	2019, eBOSS/DESI France, Cosmology with Cosmic Homogeneity (15' Talk)
	2019, Aix-University, Homogeneity with Multitracers , (2h Seminar)
	2019, Aix-University, A note on lss, Summer School , (6h Seminar)
	2018, Euclid GC-WL, Ameliorating cosmology with homogeneity , (15' Talk)
	2018, Colloquium of D.E., Cosmic Homogeneity as a standard ruler , (20' Talk)
	2018, R. Moriond, Euclid:Homogeneity in search of the Dark Sector , (Poster) HIC
	2018, UiO, What can fractality tell us about the universe? , (1h seminar)
	2017, P.O.N.T. in Cosmology, Is the universe homogeneous? , (20' Talk)
	2016, R. Moriond, The homogeneity scale of the universe , (15' Talk)
	2015, UnivEarthS School, Santorini, LSS and homogeneity , (Poster)
	2015, Rencontres de Jeunes Physicienne, Homogeneity (Video) , (15' Talk)
	2014, Elbereth Conference, IAP, The homogeneity of the universe (15' Talk)
	2014, UnivEarthS School, The homogeneity of the universe , (3' Talk)

Software

- [1] P.Ntelis (2017-today): [Github contributions](#)
- [2] P.Ntelis (2017-today): [COSMOPIT](#)
- [3] P.Ntelis (2018-2021): [Gitlab private Euclid softwares](#)
- [4] P.Ntelis (2018): [MCMC-posterior-example](#)
- [5] P.Ntelis (2018): [CoHo](#)
- [6] P.Ntelis (2018): [ANALYSE-SN-magnitude-redshift-curve](#)
- [7] P.Ntelis (2018): [Extract-Cosmology-with-the-BAO-peak-position](#)
- [8] P.Ntelis (2020): [AofEFT](#)
- [9] P.Ntelis (2023): [Tutorial on particle/galaxy simulations and coordinate transformation](#)
- [10] P.Ntelis (2025a): [Standard \$\phi\$ CDM dynamics](#)
- [11] P.Ntelis (2025b): [Simple quintessence dynamics](#)
- [12] P.Ntelis (2025c): [Analytical \$poly\Lambda\$ CDM dynamics](#)